

A class number criterion for pentads of congruent factorials

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Abstract

Let $q \equiv 1 \pmod{6}$, $q \geq 7$, and let $p \equiv 3 \pmod{4}$ be a prime divisor of $q! - 1$; such a divisor always exists, and any prime divisor of $q! - 1$ exceeds q . Building on the tetrad construction of Agustín-Aquino and Hernández Santiago, we show that the four residues $1, q, p - 1 - q, p - 2$ are pairwise distinct with all factorials $\equiv 1 \pmod{p}$, and that the central residue $m = (p - 1)/2$ joins them — producing five equal factorials, hence $k = 4$ consecutive intervals with product $\equiv 1 \pmod{p}$ in Erdős problem #1056 — *if and only if* the class number of $\mathbb{Q}(\sqrt{-p})$ satisfies $h(-p) \equiv 3 \pmod{4}$. The proof is elementary apart from the counting form of Dirichlet’s class number formula, and it recovers a theorem of Mordell. We deduce a conditional infinitude theorem for pentad primes, a multi-seed shape theorem that reduces hexad infinitude to the statement that infinitely many pairs $q_1 < q_2 \equiv 1 \pmod{6}$ satisfy $\gcd(q_1! - 1, q_2! - 1) > 1$, and a limitative theorem explaining why the classical identity toolkit terminates, unconditionally, exactly at tetrads. Twelve pentads are certified end to end; the smallest is $p = 359$ and the most striking is $p = 5039 = 7! - 1$, whose complete solution set of $n! \equiv 1 \pmod{5039}$ is precisely the constructed pentad. The unconditional infinitude question for $k \geq 4$ remains open.

1 Introduction

Erdős asked (problem #1056 in the collection at erdosproblems.com; problem A15 in Guy [4]): for $k \geq 2$, is there a prime p and k consecutive blocks

*With machine-assisted proof search and verification. All computations reported here, and substantial portions of the proof search and adversarial checking, were carried out by AI agents (Claude, Anthropic) under the author’s direction; this disclosure follows current norms for AI assistance in mathematical work. See Remark 6.1 for details on verification status.

of consecutive integers

$$I_1 = (q_0, q_1], \quad I_2 = (q_1, q_2], \quad \dots, \quad I_k = (q_{k-1}, q_k], \quad 1 \leq q_0 < q_1 < \dots < q_k \leq p-1,$$

such that the product of the elements of each block is $\equiv 1 \pmod{p}$? Since $\prod_{n \in (q_{i-1}, q_i]} n = q_i! / q_{i-1}!$, the existence of k such blocks is equivalent to the existence of $k+1$ values $q_0 < q_1 < \dots < q_k$ in $[1, p-1]$ whose factorials are congruent to a common residue modulo p . We call a set of j such values with equal factorials a j -ad: tetrads ($j = 4$) give $k = 3$, pentads ($j = 5$) give $k = 4$, hexads ($j = 6$) give $k = 5$.

Witnesses are known computationally for all $k \leq 14$: OEIS sequence A060427 [9] records the least prime for each k , beginning

11, 17, 23, 71, 71, 599, 599, 3011, 27901, 52163, 778699, 2374649, 10428007

for $k = 2, \dots, 14$ (OEIS A060427; extended by J. K. Andersen, 2007). Whether witnesses exist for *every* k , and whether any single $k \geq 2$ admits infinitely many primes, are the substantive open questions. In 2026, Agustín-Aquino and Hernández Santiago [1] proved, with a Lean-verified development, that infinitely many primes carry a tetrad: for $q \equiv 1 \pmod{6}$, $q \geq 7$, and any prime $p \mid q! - 1$, the four residues $\{1, q, p-1-q, p-2\}$ all have factorial $\equiv 1 \pmod{p}$ and are pairwise distinct. This was the state of the art: $k = 3$ with infinitely many witnesses, and nothing beyond.

This note adds one point to their configuration and finds that the question of whether it joins is governed by a class number. Our main result is the following.

Theorem 1.1 (Pentad criterion; proved as Theorem 2.1). *Let $q \equiv 1 \pmod{6}$ with $q \geq 7$, and let $p \equiv 3 \pmod{4}$ be a prime divisor of $q! - 1$. (Such a divisor always exists, since $q! - 1 \equiv 3 \pmod{4}$; and necessarily $p > q$.) Put $m = (p-1)/2$. Then the five residues*

$$\{1, q, m, p-1-q, p-2\} \subset [1, p-1]$$

are pairwise distinct, the four outer values satisfy $x! \equiv 1 \pmod{p}$, and $m! \equiv \pm 1 \pmod{p}$, with

$$m! \equiv 1 \pmod{p} \iff h(-p) \equiv 3 \pmod{4},$$

where $h(-p)$ is the class number of $\mathbb{Q}(\sqrt{-p})$. In that case the five values form a pentad and p carries $k = 4$ consecutive intervals of product $\equiv 1 \pmod{p}$; otherwise $m! \equiv -1 \pmod{p}$ and p carries the tetrad together with the pair $\{m, p-1\}$ at common value -1 .

The mechanism is a theorem of Mordell [8]: for $p \equiv 3 \pmod{4}$, $\left(\frac{p-1}{2}\right)! \equiv (-1)^{(h(-p)+1)/2} \pmod{p}$. We re-prove it here in an elementary two-step form — first $m! \equiv (-1)^{N_p} \pmod{p}$ where N_p counts quadratic nonresidues in $(0, p/2)$, then N_p even $\iff h(-p) \equiv 3 \pmod{4}$ via the counting form of Dirichlet’s class number formula — because both halves are needed separately in what follows. The point of the theorem is not the central-sign law, which is classical, but the coupling: *Erdős #1056 at $k = 4$ is tied, along an explicit infinite family of primes, to the residue of $h(-p)$ modulo 4* — a quantity about which, beyond Gauss’s parity result ($h(-p)$ is odd for prime $p \equiv 3 \pmod{4}$), no unconditional determination of $h(-p) \pmod{4}$ is known. Indeed, whether *each* residue class $h(-p) \equiv 1, 3 \pmod{4}$ occurs for infinitely many primes $p \equiv 3 \pmod{4}$ appears to be open even without any restriction to a thin family (by pigeonhole at least one class is infinite, but it is not known which); the question circulates in this form on MathOverflow since 2013, unresolved. A structural reason is supplied by Williams [10], who proved $h(-p) \equiv h(p) + U + 1 \pmod{4}$, where $T + U\sqrt{p}$ is the fundamental unit of $\mathbb{Q}(\sqrt{p})$: the invariant is governed by Pell-equation and real-quadratic data rather than by any splitting condition on p , so no Chebotarev-type criterion is available. In this sense the hypothesis of our conditional infinitude theorem (Section 4) is optimal: it isolates precisely the recognized open difficulty, with nothing else in the way.

The note is organized as follows. Section 2 states the main theorem and fixes notation; Section 3 gives complete proofs. Section 4 derives the conditional infinitude of pentad primes, an unconditional dichotomy, the multi-seed shape theorem (every $k \geq 3$ is realized by some seed pattern), and the reduction of hexad infinitude to a clean divisibility hypothesis with numerical witnesses. Section 5 proves a limitative theorem: the identity method that produces tetrads has a built-in ceiling and cannot self-start beyond them, so that pentad or hexad infinitude *via identities* is equivalent to the class number statement or to multi-seed divisibilities. Section 6 reports the numerical evidence: twelve certified pentads, the flagship $p = 5039 = 7! - 1$, and sign statistics. Remark 6.1 states plainly what has been machine-verified and what has not. We emphasize at the outset: *the tetrad construction is entirely due to Agustín-Aquino and Hernández Santiago [1]; our contribution is the central point, the class number dichotomy, and the limitative analysis. The unconditional infinitude of witnesses for any $k \geq 4$ remains open.*

2 The main theorem

Notation

Throughout, p denotes an odd prime and $m = (p-1)/2$. For a residue r we write $S_r(p) = \{n \in [1, p-1] : n! \equiv r \pmod{p}\}$. The Legendre symbol is $\left(\frac{a}{p}\right)$, and

$$N_p = \#\{a : 0 < a < p/2, \left(\frac{a}{p}\right) = -1\}, \quad A_p = \#\{a : 0 < a < p/2, \left(\frac{a}{p}\right) = +1\},$$

so $A_p + N_p = m$. We write $h(-p)$ for the class number of the imaginary quadratic field $\mathbb{Q}(\sqrt{-p})$ (equivalently, of the discriminant $-p$ when $p \equiv 3 \pmod{4}$). A *seed* for p is an integer q with $p \mid q! - 1$.

Theorem 2.1 (Pentad criterion). *Let $q \equiv 1 \pmod{6}$, $q \geq 7$, and let $p \equiv 3 \pmod{4}$ be a prime with $p \mid q! - 1$. Then:*

- (a) *such a prime p exists for every such q , and every prime divisor of $q! - 1$ satisfies $p > q$ (indeed $p \geq q + 4$);*
- (b) *the five integers $1, q, m, p-1-q, p-2$ are pairwise distinct elements of $[1, p-1]$;*
- (c) *$1! \equiv q! \equiv (p-1-q)! \equiv (p-2)! \equiv 1 \pmod{p}$, and $m! \equiv \pm 1 \pmod{p}$;*
- (d) *$m! \equiv 1 \pmod{p} \iff N_p$ is even $\iff h(-p) \equiv 3 \pmod{4}$.*

Consequently, exactly one of the following holds:

- (i) *$h(-p) \equiv 3 \pmod{4}$: then $\{1, q, m, p-1-q, p-2\}$, sorted increasingly, is a pentad at common value 1, and p carries $k = 4$ consecutive intervals each of product $\equiv 1 \pmod{p}$;*
- (ii) *$h(-p) \equiv 1 \pmod{4}$: then $m! \equiv -1 \pmod{p}$, and p carries the tetrad at +1 together with the pair $\{m, p-1\}$ at -1.*

Part (a) and the tetrad portion of (b)–(c) constitute the theorem of [1], refined by the observation — available at zero extra cost — that the prime p may always be chosen in the class 3 mod 4; the statement of [1] imposes no congruence condition on p . Parts (b) and (d) for the central point m , and the resulting dichotomy, are new to the best of our knowledge (see Remark 6.1 on the scope of the novelty check).

3 Proofs

3.1 The reflection lemma

The engine of every construction in this subject is the Wilson pairing identity. It is classical; we state it with the consequences we use.

Lemma 3.1 (Reflection). *Let p be an odd prime and $0 \leq n \leq p-1$. Then*

$$n!(p-1-n)! \equiv (-1)^{n+1} \pmod{p}.$$

Consequently:

- (i) *if n is odd and $n! \equiv c \pmod{p}$, then $(p-1-n)! \equiv c^{-1} \pmod{p}$; in particular the sets $S_1(p)$ and $S_{-1}(p)$ are stable under $n \mapsto p-1-n$ restricted to odd n ;*
- (ii) *if n is even and $n! \equiv 1$, then $(p-1-n)! \equiv -1$ (and vice versa);*
- (iii) *$1, p-2 \in S_1(p)$ and $p-1 \in S_{-1}(p)$ for $p \geq 5$;*
- (iv) *$(p-k)! \equiv (-1)^k ((k-1)!)^{-1} \pmod{p}$ for $1 \leq k \leq p-1$; hence $(p-k)! \equiv 1$ is equivalent to $p \mid (k-1)! - (-1)^k$.*

Proof. Write $(p-1)! = n! \prod_{j=n+1}^{p-1} j$. For $n+1 \leq j \leq p-1$ put $i = p-j$, so $1 \leq i \leq p-1-n$ and $j \equiv -i \pmod{p}$; hence

$$\prod_{j=n+1}^{p-1} j \equiv (-1)^{p-1-n} (p-1-n)! \equiv (-1)^n (p-1-n)! \pmod{p},$$

since $p-1$ is even. Wilson's theorem $(p-1)! \equiv -1$ gives the identity. (i): n odd makes the right side $+1$, so $(p-1-n)! \equiv (n!)^{-1} = c^{-1}$; since $c = \pm 1$ are the only self-inverse residues in $\mathbb{F}_p$, exactly the classes $S_{\pm 1}$ are preserved (the class of $c \neq \pm 1$ is carried to the class of $c^{-1} \neq c$). (ii): n even makes the right side -1 , so $(p-1-n)! \equiv -c^{-1}$. (iii): $1! = 1$; from $(p-1)! = (p-1)(p-2)! \equiv -(p-2)! \equiv -1$ we get $(p-2)! \equiv 1$; and $(p-1)! \equiv -1$ is Wilson. (iv): $(p-1)! = (p-1)(p-2) \cdots (p-(k-1)) \cdot (p-k)! \equiv (-1)^{k-1} (k-1)! (p-k)! \equiv -1$, so $(p-k)! \equiv (-1)^k ((k-1)!)^{-1}$. $\square$

3.2 Existence of the seed prime, and its size

Lemma 3.2 (Seed prime). *For every integer $q \geq 4$, the number $q! - 1$ satisfies $q! - 1 \equiv 3 \pmod{4}$, and hence has at least one prime factor $p \equiv 3 \pmod{4}$. Moreover every prime factor of $q! - 1$ exceeds q .*

Proof. Since $q \geq 4$, $4 \mid q!$, so $q! - 1 \equiv 3 \pmod{4}$. An odd integer all of whose prime factors are $\equiv 1 \pmod{4}$ is itself $\equiv 1 \pmod{4}$, because the class of 1 is closed under multiplication in $(\mathbb{Z}/4\mathbb{Z})^\times$; hence $q! - 1$ has a prime factor $p \equiv 3 \pmod{4}$. If a prime $p \leq q$ divided $q! - 1$, it would divide $q!$ and hence their difference 1, which is absurd; so $p > q$. $\square$

Remark 3.3. The lemma cannot be sharpened to a class modulo 8: indeed $q! - 1 \equiv 7 \pmod{8}$ for $q \geq 4$, but $3 \cdot 5 \equiv 7 \pmod{8}$ shows that an integer $\equiv 7 \pmod{8}$ need not have any prime factor $\equiv 7 \pmod{8}$. The modulo-4 conclusion is exactly what Mordell's theorem needs, and exactly what is available.

3.3 Distinctness

Lemma 3.4 (Distinctness). *Let $q \equiv 1 \pmod{6}$, $q \geq 7$, and let $p > q$ be an odd prime, $m = (p-1)/2$. Then the five integers $1, q, m, p-1-q, p-2$ are pairwise distinct elements of $[1, p-1]$, and moreover $p \geq q+4$.*

Proof. The pairwise coincidences among the four tetrad values reduce to six equations:

$$\begin{aligned} 1 = q; \quad 1 = p-1-q &\iff p = q+2; \quad 1 = p-2 \iff p = 3; \\ q = p-1-q &\iff p = 2q+1; \quad q = p-2 \iff p = q+2; \quad p-1-q = p-2 \iff q = 1. \end{aligned}$$

Adjoining m adds four more:

$$m = 1 \iff p = 3; \quad m = p-2 \iff p = 3; \quad m = q \iff p = 2q+1; \quad m = p-1-q \iff p = 2q+1.$$

Now $q \equiv 1 \pmod{6}$ forces $q+2 \equiv 3 \pmod{6}$ and $2q+1 \equiv 3 \pmod{6}$; both exceed 3, so both are divisible by 3 and composite — neither can equal the prime p . Also $p > q \geq 7 > 3$ excludes $p = 3$, and $q = 1$ is excluded by hypothesis. Hence no coincidence occurs. Finally, p is odd and $p \notin \{q+1, q+3\}$ (both even) and $p \neq q+2$, so $p \geq q+4$; in particular $p-1-q \geq 3 \geq 1$ and all five values lie in $[1, p-1]$. $\square$

3.4 The central sign law

Lemma 3.5 (Central sign). *Let $p \equiv 3 \pmod{4}$ be prime, $p > 3$, and $m = (p-1)/2$ (which is odd). Then*

$$m! \equiv (-1)^{N_p} \pmod{p}.$$

Proof. Let $R \subset \mathbb{F}_p^\times$ be the set of quadratic residues, a subgroup of odd order $|R| = m$. In a finite abelian group of odd order the product of all elements is the identity: elements pair off with their inverses, and no element other than the identity is self-inverse. Hence $\prod_{r \in R} r \equiv 1 \pmod{p}$.

Since $p \equiv 3 \pmod{4}$ we have $\left(\frac{-1}{p}\right) = -1$, so the map $a \mapsto p - a$ exchanges residues and nonresidues; in particular it maps the nonresidues in $(0, p/2)$ bijectively onto the residues in $(p/2, p)$. Therefore

$$R = (R \cap (0, p/2)) \sqcup \{p - a : 0 < a < p/2, \left(\frac{a}{p}\right) = -1\},$$

and consequently

$$1 \equiv \prod_{r \in R} r \equiv \left(\prod_{\substack{0 < a < p/2 \\ \left(\frac{a}{p}\right) = 1}} a \right) \prod_{\substack{0 < a < p/2 \\ \left(\frac{a}{p}\right) = -1}} (p - a) \equiv (-1)^{N_p} \prod_{0 < a < p/2} a = (-1)^{N_p} m! \pmod{p}.$$

Since $(-1)^{N_p}$ is its own inverse, $m! \equiv (-1)^{N_p} \pmod{p}$. (Consistently, Lemma 3.1 at the odd fixed point $n = m$ gives $(m!)^2 \equiv 1$.) $\square$

3.5 The class number bridge

Lemma 3.6 (Class number bridge). *Let $p \equiv 3 \pmod{4}$ be prime, $p > 3$. Then $h(-p)$ is odd, and*

$$N_p \text{ is even} \iff h(-p) \equiv 3 \pmod{4}.$$

Combined with Lemma 3.5 this gives Mordell's theorem [8]:

$$m! \equiv (-1)^{(h(-p)+1)/2} \pmod{p}, \quad \text{i.e.} \quad m! \equiv +1 \iff h(-p) \equiv 3 \pmod{4}.$$

Proof. Oddness of $h(-p)$ is Gauss's genus theory: the discriminant $-p$ has a single prime discriminant divisor, so there is one genus, and the number of classes per genus is odd; equivalently, the 2-part of the class group is trivial.

For the mod-4 statement we use the counting form of Dirichlet's class number formula for a prime discriminant $-p < -4$ (see Landau [7] or Borevich–Shafarevich [2]):

$$h(-p) = \frac{A_p - N_p}{2 - \left(\frac{2}{p}\right)}.$$

Recall $A_p + N_p = m$. There are two branches.

Case $p \equiv 7 \pmod{8}$. Then $\left(\frac{2}{p}\right) = +1$ and $h = A_p - N_p = m - 2N_p$. Writing $p = 8s + 7$ gives $m = 4s + 3 \equiv 3 \pmod{4}$, so $h \equiv 3 - 2N_p \pmod{4}$: $h \equiv 3 \pmod{4}$ iff $2N_p \equiv 0 \pmod{4}$ iff N_p is even.

Case $p \equiv 3 \pmod{8}$. Then $\left(\frac{2}{p}\right) = -1$ and $3h = m - 2N_p$. Writing $p = 8s + 3$ gives $m = 4s + 1 \equiv 1 \pmod{4}$, so $3h \equiv 1 - 2N_p \pmod{4}$ and, multiplying by $3^{-1} \equiv 3 \pmod{4}$,

$$h \equiv 3(1 - 2N_p) \equiv 3 + 2N_p \pmod{4} :$$

again $h \equiv 3 \pmod{4}$ iff N_p is even.

In both branches N_p even $\iff h(-p) \equiv 3 \pmod{4}$. Since h is odd, $(-1)^{(h+1)/2} = +1 \iff h \equiv 3 \pmod{4}$, which matches $m! \equiv (-1)^{N_p}$ from Lemma 3.5. $\square$

Remark 3.7. In our numerical verification (Section 6) the class numbers were computed by exact counting of reduced primitive binary quadratic forms of discriminant $-p$, independently cross-checked against the Dirichlet character sum; the verification of the criterion therefore does not assume the analytic formula it uses.

3.6 Proof of Theorem 2.1

Proof of Theorem 2.1. (a) is Lemma 3.2 (with $p \geq q + 4$ from Lemma 3.4); (b) is Lemma 3.4.

(c): $1! = 1$; $q! \equiv 1$ because $p \mid q! - 1$; q is odd, so Lemma 3.1(i) gives $(p - 1 - q)! \equiv 1$; and $(p - 2)! \equiv 1$ by Lemma 3.1(iii). Since $p \equiv 3 \pmod{4}$, m is odd, and Lemma 3.1 at $n = m$ gives $(m!)^2 \equiv (-1)^{m+1} = 1$, so $m! \equiv \pm 1 \pmod{p}$.

(d) is Lemma 3.5 combined with Lemma 3.6.

For the dichotomy: in case (i) the five distinct values, sorted as $q_0 < q_1 < \dots < q_4$, all lie in $S_1(p)$, so each consecutive interval $(q_{i-1}, q_i]$ has product $q_i! / q_{i-1}! \equiv 1 \pmod{p}$ — four intervals, i.e. $k = 4$ in the original formulation. In case (ii), $m! \equiv -1$ places m with the Wilson point $p - 1$ in $S_{-1}(p)$. The two cases are exhaustive and exclusive by (c) and (d). $\square$

4 Corollaries: conditional infinitude, shapes, and hexads

4.1 Pentad primes, conditionally

Let

$$\mathcal{F} = \{ (q, p) : q \equiv 1 \pmod{6}, q \geq 7, p \text{ prime}, p \mid q! - 1, p \equiv 3 \pmod{4} \}.$$

By Lemma 3.2, $\mathcal{F}$ projects onto all admissible q , and $p > q$ throughout.

Hypothesis 4.1 (CFS: central factorial sign). $h(-p) \equiv 3 \pmod{4}$ for infinitely many pairs $(q, p) \in \mathcal{F}$.

Full equidistribution — each residue $1, 3 \pmod{4}$ with density $1/2$ — is what randomness models for class numbers predict, and what the data of Section 6 show; but only “infinitely often” is needed.

Corollary 4.2 (Conditional infinitude of pentad primes). *Assume Hypothesis 4.1. Then infinitely many primes p admit five distinct values $x \in [1, p-1]$ with $x! \equiv 1 \pmod{p}$; that is, Erdős #1056 holds for $k = 4$ with infinitely many prime witnesses.*

Proof. For each pair $(q, p) \in \mathcal{F}$ with $h(-p) \equiv 3 \pmod{4}$, Theorem 2.1(i) produces a pentad for p . It remains to see the primes involved are unbounded. For fixed q the number $q! - 1$ has finitely many prime factors, so an infinite set of pairs contains pairs with q arbitrarily large; and $p > q$ forces the corresponding primes to be unbounded, hence infinitely many distinct primes occur. $\square$

Corollary 4.3 (Unconditional dichotomy). *At least one of the following is true:*

- (i) *infinitely many primes admit pentads at value 1 ($k = 4$ with infinitely many witnesses); or*
- (ii) *$h(-p) \equiv 1 \pmod{4}$ for all but finitely many pairs $(q, p) \in \mathcal{F}$ — an eventually-constant law for the residue of an odd class number along an explicit unbounded family of primes, contrary to every randomness model for $h(-p)$.*

Proof. If case (i) of Theorem 2.1 occurs for infinitely many members of $\mathcal{F}$, the argument of Corollary 4.2 gives conclusion (i); if it occurs for only finitely many, conclusion (ii) holds by definition. $\square$

4.2 The multi-seed shape theorem

The pentad construction generalizes: each additional seed buys more points, with all degeneracies killed automatically by the mod-6 arithmetic.

Theorem 4.4 (Shape theorem). *Let $j \geq 1$ and $e \geq 0$. Choose distinct integers $q_1 < \dots < q_j$, each $\equiv 1 \pmod{6}$ and ≥ 7 (hence odd), and distinct even integers $t_1, \dots, t_e \geq 4$ (no congruence condition needed). Suppose a prime p divides $q_i! - 1$ for every $i \in \{1, \dots, j\}$ and $t_\ell! - 1$ for every $\ell \in \{1, \dots, e\}$. Then p exceeds every seed, and the $2j + e + 2$ integers*

$$\{1, p-2\} \cup \{q_1, \dots, q_j\} \cup \{t_1, \dots, t_e\} \cup \{p-1-q_1, \dots, p-1-q_j\}$$

are pairwise distinct elements of $[1, p-1]$ with all factorials $\equiv 1 \pmod{p}$. Sorted increasingly, they yield $k = 2j + e + 1$ consecutive intervals each of product $\equiv 1 \pmod{p}$.

Value-(-1) variant: if instead $p \mid q_i! + 1$ and $p \mid t_i! + 1$, then $\{p-1\} \cup \{q_i\} \cup \{t_i\} \cup \{p-1-q_i\}$ are $2j + e + 1$ distinct points with all factorials $\equiv -1$, yielding $k = 2j + e$ intervals.

In particular $(j, e) = (1, 0)$ recovers the tetrad theorem of [1] ($k = 3$); $(1, 1)$ gives pentads ($k = 4$); $(2, 0)$ gives hexads ($k = 5$); and every $k \geq 3$ (indeed every $k \geq 2$, using the -1 variant) is realized by some (j, e) .

Proof. Size of p : if $p \leq q$ for some seed q , then $p \mid q!$ and $p \mid q! \mp 1$ force $p \mid 1$, absurd; so p exceeds every seed, and since p is odd, $p \geq (\text{largest odd seed}) + 2$, so all listed points lie in $[1, p-1]$.

Factorial values (+1 case): $1! = 1$; $q_i! \equiv 1$ and $t_i! \equiv 1$ by hypothesis; $(p-2)! \equiv 1$ by Lemma 3.1(iii); $(p-1-q_i)! \equiv 1$ by Lemma 3.1(i), since q_i is odd.

Distinctness — the complete case list:

- [a] $1 = q_i$ or $1 = t_i$: impossible, all seeds are ≥ 4 .
- [b] $1 = p-2 \iff p = 3$: impossible, p exceeds the seeds.
- [c] $1 = p-1-q_i \iff p = q_i + 2$: but $q_i \equiv 1 \pmod{6}$ makes $q_i + 2 \equiv 3 \pmod{6}$, so $3 \mid p$, forcing $p = 3$: impossible.
- [d] $q_i = q_{i'}$ or $t_i = t_{i'}$ ($i \neq i'$): excluded by choice.
- [e] $q_i = t_{i'}$: opposite parities.
- [f] $q_i = p-2 \iff p = q_i + 2$: as [c].

- [g] $t_i = p - 2 \iff p = t_i + 2$, which is even: impossible for odd p .
- [h] $q_i = p - 1 - q_{i'} \iff p = q_i + q_{i'} + 1$: for $i = i'$ this is $p = 2q_i + 1 \equiv 3 \pmod{6}$; for $i \neq i'$ it is $\equiv 1 + 1 + 1 \equiv 3 \pmod{6}$; either way $3 \mid p$: impossible.
- [i] $t_i = p - 1 - q_{i'} \iff p = q_{i'} + t_i + 1 = \text{odd} + \text{even} + 1$, which is even: impossible.
- [j] $p - 1 - q_i = p - 1 - q_{i'} \iff q_i = q_{i'}$.
- [k] $p - 1 - q_i = p - 2 \iff q_i = 1$: impossible.

So all $2j + e + 2$ points are distinct, and consecutive interval products are ratios of equal factorials, hence $\equiv 1$.

(-1) *variant*: values: $(p - 1)! \equiv -1$ (Wilson); $q_i! \equiv t_i! \equiv -1$ by hypothesis; $(p - 1 - q_i)! \equiv (-1)^{-1} = -1$ by Lemma 3.1(i). Distinctness: cases [d], [e], [h]–[j] as above; $q_i = p - 1$ or $t_i = p - 1$ would make $p = \text{seed} + 1$ even: impossible; $p - 1 - q_i = p - 1 \iff q_i = 0$. All cases close, giving $2j + e + 1$ distinct points and $k = 2j + e$ intervals.

Realization of every k : $k = 2j + e + 1$ with $j \geq 1$, $e \in \{0, 1\}$ covers all $k \geq 3$; the -1 variant covers $k \geq 2$. A witness for k also witnesses every $k' < k$ (take a subset of the points). $\square$

Note that Theorem 4.4 makes no claim that primes satisfying several simultaneous seed divisibilities *exist* — that is precisely the hard part; see Section 5. What it shows is that *whenever* they exist, the resulting configuration is automatically nondegenerate.

4.3 Hexad infinitude reduced to a divisibility statement

Corollary 4.5 (Hexad reduction). *If there are infinitely many pairs $q_1 < q_2$ with $q_1 \equiv q_2 \equiv 1 \pmod{6}$, $q_1 \geq 7$, and $\gcd(q_1! - 1, q_2! - 1) > 1$, then infinitely many primes admit hexads at value 1, i.e. Erdős #1056 holds for $k = 5$ (hence also $k = 3, 4$) with infinitely many prime witnesses.*

No size or congruence condition on the shared prime is needed: every prime factor of $\gcd(q_1! - 1, q_2! - 1)$ automatically exceeds q_2 and automatically avoids all degenerate collisions.

Proof. Let (q_1, q_2) be such a pair and p any prime factor of the gcd. By Theorem 4.4 with $(j, e) = (2, 0)$, $p > q_2$ and p carries the hexad $\{1, q_1, q_2, p - 1 - q_2, p - 1 - q_1, p - 2\}$. Given infinitely many pairs: for fixed q_2 there are fewer than q_2 pairs, so q_2 takes arbitrarily large values along the family; each

pair's prime satisfies $p > q_2$, so the set of primes produced is unbounded, hence infinite, and each carries $k = 5$. $\square$

The hypothesis is non-vacuous: nontrivial factorial gcd pairs within the theorem-exact family exist in abundance. Verified witnesses $(q_1, q_2; p)$ with $q_1 \equiv q_2 \equiv 1 \pmod{6}$ and $p \mid \gcd(q_1! - 1, q_2! - 1)$ include

$$(49, 97; 823), \quad (13, 361; 1733), \quad (415, 841; 2287), \quad (193, 265; 2113), \\ (361, 505; 971), \quad (445, 1045; 5477);$$

on the -1 side, $p = 71$ divides $q! + 1$ simultaneously for the seeds $q = 7, 19, 61$ (all $\equiv 1 \pmod{6}$; indeed $7! + 1 = 71^2$), whose -1 -variant configuration realizes $S_{-1}(71) = \{7, 9, 19, 51, 61, 63, 70\}$, a heptad at -1 . Note also that $p = 971$ appears both as a pentad prime (seed 361, Table 1) and as a hexad prime (seeds 361, 505): the mechanisms stack. A heuristic count (independent-values model, corrected for the Wilson seed $q = p-2$ and reflection doubling) predicts that the number of such pairs with $q_2 \leq Q$ grows like $Q/\log Q$ up to constants, and measured counts up to $Q = 1500$ grow near-linearly as predicted; but no proof technique for the infinitude of such pairs is currently visible.

5 Limits of the identity method

The results above raise an obvious question: the tetrad theorem is unconditional — why should the pentad cost a class number condition, and the hexad a simultaneous-divisibility hypothesis? This section proves that the barrier is structural, not an artifact of our construction. The “identity method” below means: producing members of $S_1(p)$ for infinitely many p using the universal identities of Lemma 3.1 together with primes constructed as divisors of integer sequences (Euclid-style seeds $p \mid q! \mp 1$).

5.1 Template rigidity

Proposition 5.1 (Template rigidity). *Call ν a template position if $\nu(p) = a$ (fixed $a \geq 0$), or $\nu(p) = p - j$ (fixed $j \geq 1$), or $\nu(p) = m \pm t$ (fixed $t \geq 0$). Then $\nu(p)! \equiv 1 \pmod{p}$ holds for infinitely many primes p only in the cases*

$$a \in \{0, 1\}, \quad j = 2, \quad (t = 0 \text{ with } p \equiv 3 \pmod{4} \text{ and } h(-p) \equiv 3 \pmod{4}).$$

In every other case the set of admissible p is finite and explicitly contained in the divisors of a fixed nonzero integer:

$$a \geq 2 : p \mid a! - 1; \quad j \notin \{1, 2\}, \quad p > j : p \mid (j-1)! - (-1)^j;$$

$$t \geq 1, p \equiv 3 \pmod{4} : p \mid ((2t-1)!!)^2 - 4^t; \quad t \geq 1, p \equiv 1 \pmod{4} : p \mid ((2t-1)!!)^2 + 4^t.$$

Proof sketch. Fixed low positions $a \geq 2$ are immediate. Top positions: by Lemma 3.1(iv), $(p-j)! \equiv 1 \iff p \mid (j-1)! - (-1)^j$, a nonzero integer for $j \notin \{1, 2\}$ since $(j-1)! \geq 2$ ($j=1$ gives $p \mid 2$; $j=2$ gives 0, i.e. all p). Near-center positions: the telescoped central formulas

$$(m+t)! \equiv m! \cdot \frac{(2t-1)!!}{2^t}, \quad (m-t)! \equiv m! \cdot \frac{(-2)^t}{(2t-1)!!} \pmod{p}$$

(both consequences of $j \equiv -(p-j)$ pairing around the center) combine with $(m!)^2 \equiv (-1)^{m+1}$: for $p \equiv 3 \pmod{4}$, $m! = \sigma \in \{\pm 1\}$ and $(m \pm t)! \equiv 1$ forces $p \mid (2t-1)!! \mp \sigma(\pm 2)^t$, hence in either case $p \mid ((2t-1)!!)^2 - 4^t$, nonzero (odd square minus even power) for $t \geq 1$; for $p \equiv 1 \pmod{4}$, $(m!)^2 \equiv -1$ and squaring gives $p \mid ((2t-1)!!)^2 + 4^t$. The center itself ($t=0$): for $p \equiv 1 \pmod{4}$, $(m!)^2 \equiv -1$, so $m!$ has order 4 and is never ± 1 ; for $p \equiv 3 \pmod{4}$ the sign is $h(-p) \pmod{4}$ (Lemmas 3.5–3.6), which admits no known congruence description: empirically both signs occur with frequency $\frac{1}{2} \pm 0.03$ in every residue class modulo 8, 16, and 24 for $p \leq 60000$. $\square$

5.2 Bounded patterns are finite

A *pattern* is a composition $(L_1, \dots, L_t)$ of a total length $C = L_1 + \dots + L_t$, describing t adjacent intervals of consecutive integers of lengths L_i , placed at a variable base: with cut points $c_0 = 0 < c_1 < \dots < c_t = C$ and base m , the chain condition $(m+c_0)! \equiv (m+c_1)! \equiv \dots \equiv (m+c_t)! \pmod{p}$ (for $p > m+C$) is equivalent to the polynomial system

$$g_i(m) \equiv 0 \pmod{p}, \quad g_i(x) = \prod_{j=c_{i-1}+1}^{c_i} (x+j) - 1, \quad i = 1, \dots, t.$$

Theorem 5.2 (Bounded-pattern finiteness; computational). *For every one of the 65,519 patterns with $C \leq 16$ and $t \geq 2$, and every one of the 1,966,080 resulting polynomial pairs (g_i, g_j) , $\gcd(g_i, g_j) = 1$ in $\mathbb{Q}[x]$. Consequently, for every such pattern the primes admitting a chain with all cut points inside a window of length ≤ 16 divide the (nonzero) pairwise resultants $\text{Res}(g_i, g_j)$ — a finite, explicit set per pattern. There is no bounded-pattern route to infinitude for any $k \geq 2$, and no “algebraic jackpot” pattern exists at small size.*

Proof sketch. The equivalence chain-condition $\leftrightarrow$ polynomial system follows from $(m+c_i)! = (m+c_{i-1})! \prod_{j=c_{i-1}+1}^{c_i} (m+j)$ and $p \nmid (m+c_{i-1})!$.

For the coprimality certificate: reducing modulo a prime ℓ can only increase the degree of a gcd (Sylvester/subresultant argument), so exhibiting $\deg \gcd_{\mathbb{F}_\ell}(g_i, g_j) = 0$ for a single ℓ proves coprimality over $\mathbb{Q}$; the computation used $\ell = 2^{61} - 1$ for every pair. If $\gcd(g_i, g_j) = 1$ then $\text{Res}(g_i, g_j) \neq 0$, and any prime modulo which g_i, g_j share a root divides it. $\square$

The historically known sporadic witnesses are precisely such resultant primes: for the pattern $(2, 3)$, $\text{Res}(g_1, g_2) = 11$ exactly — Erdős's example $3 \cdot 4 \equiv 5 \cdot 6 \cdot 7 \equiv 1 \pmod{11}$ lives at the root $m \equiv 2 \pmod{11}$ — and 17 divides all three pairwise resultants of Makowski's pattern $(4, 6, 4)$, root $m \equiv 1 \pmod{17}$. In this light, A060427 is a record sequence drawn from prime factors of resultant systems of provably coprime polynomial families, plus unstructured coincidences: sporadic by nature. For $C > 16$ pairwise coprimality remains conjectural; it is a close cousin of the conjecture of Klurman and Munsch [6] (Conjecture 3.3 in the published version; Conjecture 4.2 in arXiv:1505.01198) that the splitting fields of $t(t+1) \cdots (t+n-1) - 1$ for distinct n intersect only in $\mathbb{Q}$ — that is, the experts' conjecture is that this obstruction is *permanent*.

5.3 Cheap second divisibilities do not exist

Lemma 5.3. *For all integers $q, n \geq 2$:*

$$\gcd(q!-1, (q+1)!-1) = 1; \quad \gcd(q!-1, q!+1) = 1; \quad \gcd(n!+1, (n+1)!+1) = 1;$$

and $\gcd(q!-1, (q+1)!+1)$ divides $q+2$, hence equals 1 for $q \equiv 1 \pmod{6}$, $q \geq 7$.

Proof. If $d \mid q!-1$ and $d \mid (q+1)!-1$ then $d \mid (q+1)(q!-1) - ((q+1)!-1) = -q$, so $d \mid q$; but $d \mid q!-1$ forces $\gcd(d, q!) = 1$, so $d = 1$. The second gcd divides 2 and both entries are odd. The third: $d \mid (n+1)(n!+1) - ((n+1)!+1) = n$, and $\gcd(d, n!) = 1$ again. The fourth: $d \mid (q+1)!+1 - (q+1)(q!-1) = q+2$; for $q \equiv 1 \pmod{6}$ every prime factor of $q+2 = 3 \cdot \frac{q+2}{3}$ is $\leq \frac{q+2}{3} < q$ or equals 3, while every prime factor of $q!-1$ exceeds q ; so $d = 1$. $\square$

5.4 The bootstrap ceiling

Proposition 5.4 (Bootstrap). *Fix $t \geq 1$ and odd seeds $q_1 < \cdots < q_t$. The t simultaneous conditions $q_i! \equiv 1 \pmod{p}$ are equivalent to*

$$p \mid q_1! - 1 \quad \text{together with} \quad p \mid \prod_{q_{i-1} < j \leq q_i} j - 1, \quad i = 2, \dots, t$$

— that is, to a witness of the anchored t -interval instance of Erdős #1056 itself. Hence the identity method converts infinitude for the t -seed configuration ($k = 2t + 1$ intervals by Theorem 4.4) into infinitude for a t -interval instance of the same problem, and cannot self-start beyond $t = 1$: the case $t = 1$ is exactly the single-sequence Euclid construction (the tetrad theorem of [1], $k = 3$), while every $t \geq 2$ presupposes an instance of the problem being solved.

Proof. $p \mid q_1! - 1$ implies $p \nmid q_i!$ for all i , so $q_i! \equiv 1 \iff$ the interval product over $(q_{i-1}, q_i]$ is $\equiv 1$; conversely those interval conditions are precisely “ $t - 1$ further consecutive intervals of product $\equiv 1$, anchored at $q_1 \in S_1(p)$ ”. $\square$

5.5 The exhaustion theorem and the equivalence

Theorem 5.5 (Exhaustion). *Fix $W \geq 3$ and let $C(W)$ be the maximum of the finite set*

$$\{n! - 1 : 2 \leq n \leq W\} \cup \{|(j-1)! - (-1)^j| : 3 \leq j \leq W\} \cup \{|((2t-1)!!)^2 \mp 4^t| : 1 \leq t \leq W\}.$$

Then for every prime $p > C(W)$, with $m = (p - 1)/2$:

$$S_1(p) \cap \left([1, W] \cup [p-W, p-1] \cup [m-W, m+W] \right) = \{1, p-2\} \cup \begin{cases} \{m\} & \text{if } p \equiv 3 \pmod{4} \text{ and } h(-p) \equiv 3 \\ \emptyset & \text{otherwise.} \end{cases}$$

Proof sketch. Each window is handled by Proposition 5.1: a member of $S_1(p)$ at position $n \in [2, W]$, $n = p - j$ with $3 \leq j \leq W$, or $n = m \pm t$ with $1 \leq t \leq W$ forces p to divide a nonzero integer bounded by $C(W) < p$, which is impossible; the surviving positions are exactly 1, $p - 2$, and the conditional center. $\square$

Assembling Proposition 5.1, Theorem 5.2, Lemma 5.3, Proposition 5.4, and Theorem 5.5:

Theorem 5.6 (Limitative theorem). *Within the identity framework — members of $S_1(p)$ produced by the universal identities of Lemma 3.1 at template positions, with primes obtained as divisors of constructed integers — the following hold.*

- (i) *One divisibility $p \mid q! - 1$ (q odd) yields exactly the orbit $\{q, p - 1 - q\}$ of the Wilson involution, on top of the universal pair $\{1, p - 2\}$: four elements, the tetrad, $k = 3$. This is the unconditional ceiling of the method.*

- (ii) *The fifth element requires the center, i.e. the open statement $h(-p) \equiv 3 \pmod{4}$ infinitely often along $\mathcal{F}$ (Hypothesis 4.1); there is no congruence handle on this sign.*
- (iii) *The sixth element requires a second independent divisibility $p \mid \gcd(q_1! - 1, q_2! - 1)$; cheap adjacent variants are provably trivial (Lemma 5.3), bounded patterns admit only finitely many primes each (Theorem 5.2), and unbounded second seeds re-pose the problem itself (Proposition 5.4).*

Hence pentad or hexad infinitude via identities is equivalent to the class number statement (Hypothesis 4.1) or to the multi-seed divisibility statement (hypothesis of Corollary 4.5), respectively. This explains precisely why the June 2026 state of the art ($k = 3$) is the natural ceiling of the entire classical toolkit.

We regard Theorem 5.6 as the real content of this note alongside Theorem 2.1: not a proof that $k \geq 4$ infinitude is false — we believe the opposite — but an identification of the exact open statements any identity-based proof must pass through.

6 Numerical evidence

6.1 Certified pentads

Table 1 lists twelve pentads certified end to end: the seed divisibility $p \mid q! - 1$, the primality of p , the congruences $p \equiv 3 \pmod{4}$ and $q \equiv 1 \pmod{6}$, the five factorial values, distinctness, and the class number $h(-p) \equiv 3 \pmod{4}$ were each verified by direct computation, with $h(-p)$ computed two independent ways (exact count of reduced primitive binary quadratic forms of discriminant $-p$, and the Dirichlet character sum), in agreement in every case. Note that composite seeds are allowed and occur ($q = 25 = 5^2$, $q = 115$, $q = 235$, $q = 295$, $q = 301$, $q = 361 = 19^2$).

6.2 The flagship

The prime $p = 5039 = 7! - 1$ is itself of the form $q! - 1$ with the seed $q = 7 \equiv 1 \pmod{6}$, and $5039 \equiv 3 \pmod{4}$ with $h(-5039) = 83 \equiv 3 \pmod{4}$. An exhaustive scan of $n = 1, \dots, 5038$ shows that the complete solution set of $n! \equiv 1 \pmod{5039}$ is *exactly* the constructed pentad

$$\{1, 7, 2519, 5031, 5037\},$$

p	q	$m = \frac{p-1}{2}$	$p-1-q$	$h(-p)$	
359	229	179	129	19	
439	37	219	401	15	
883	43	441	839	3	
907	25	453	881	3	
971	361	485	609	15	
5003	295	2501	4707	15	
5039	7	2519	5031	83	(flagship: $5039 = 7! - 1$)
9439	235	4719	9203	75	
9643	337	4821	9305	11	
9791	301	4895	9489	119	
21839	115	10919	21723	195	
22639	223	11319	22415	55	

Table 1: Twelve certified pentads. In each row the pentad is $\{1, q, m, p-1-q, p-2\}$, all five factorials $\equiv 1 \pmod{p}$, and $h(-p) \equiv 3 \pmod{4}$ as the criterion requires.

and $n! \equiv -1 \pmod{5039}$ only at the Wilson point $n = 5038$. Thus for this prime the theorem does not merely produce five solutions; it accounts for every solution.

6.3 Tightness in the negative direction

The criterion bites both ways. For $q = 31$, the divisor $p = 787$ of $31! - 1$ has $h(-787) \equiv 1 \pmod{4}$; for $q = 13$, the only prime factor $\equiv 3 \pmod{4}$ of $13! - 1$ is $p = 3593203$, with $h(-p) \equiv 1 \pmod{4}$. In both cases $m! \equiv -1 \pmod{p}$ was verified directly, and the scan confirmed that no pentad through m exists: case (ii) of Theorem 2.1, exactly as predicted.

6.4 Sign statistics

Across all 1632 primes $p \equiv 3 \pmod{4}$ in $[7, 30000]$, the central sign $m! \equiv \pm 1 \pmod{p}$ splits 826 at $+1$ to 806 at -1 — consistent with the expected equidistribution of $h(-p) \pmod{4}$ between 1 and 3. The split remains balanced (within ± 0.03 of $\frac{1}{2}$) when refined by residue classes modulo 8, 16, and 24 for $p \leq 60000$; no congruence bias is visible, which is why Hypothesis 4.1 appears both true and inaccessible to congruence methods.

6.5 Independent verification

The elementary components were stress-tested independently of the proofs: the reflection identity of Lemma 3.1 was verified exhaustively for all $n \in [0, p - 1]$ for every prime $p < 500$ and for $p = 10007$ and $p = 99991$; the Mordell law (Lemmas 3.5–3.6) was verified for all 1135 primes $p \equiv 3 \pmod{4}$ in $[7, 20000)$, with class numbers computed by reduced-form counting — millions of individual assertions in total, with zero failures.

Remark 6.1 (Provenance, verification status, and honesty). This note reports work carried out in an AI-assisted setting: the proof search was conducted by multiple independent prover agents, the results were attacked by adversarial refuter agents, and all computations described above were machine-executed and machine-checked. The reader should weigh the following plainly stated caveats. (1) The elementary core (Lemmas 3.1–3.6 and Theorem 2.1) is short enough to verify by hand, and a Lean formalization of this core is in progress but not complete at the time of writing. (2) The novelty assessment — that the coupling of Erdős #1056 to $h(-p) \pmod{4}$ does not appear in the tetrads development [1], in the discussion at erdosproblems.com/1056 [3], or in Hardy–Subbarao [5] — rests on machine literature searches and a dedicated follow-up check, not on a systematic human review; the coupling is natural and prior art may exist. (3) The Wilson pairing toolkit itself is classical and no novelty is claimed for it, nor for Mordell’s theorem, of which Lemmas 3.5 and 3.6 are a re-proof. (4) The tetrad construction and the family $q \equiv 1 \pmod{6}$, $p \mid q! - 1$ are entirely due to Agustín-Aquino and Hernández Santiago [1]; our theorem adds the central point and the class number dichotomy. (5) Above all: *no unconditional infinitude for any $k \geq 4$ is claimed*. Corollary 4.2 is conditional on Hypothesis 4.1, Corollary 4.5 on a divisibility hypothesis, and Theorem 5.6 explains why some such hypothesis is currently unavoidable. The Erdős problem remains open.

Acknowledgments

The tetrad theorem and its Lean development by O. A. Agustín-Aquino and J. Hernández Santiago [1] are the foundation on which this note builds. Thanks are due to the maintainers of erdosproblems.com and the OEIS, whose curation makes attacks like this one possible.

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